WORKED SOLUTION

STEP II 2011, Question 3

Pure • Inequalities via Monotonicity

STEP II 2011, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q3

The key idea

Every inequality here comes from one principle: if a function starts at zero and has a non-negative derivative, it stays non-negative. The last two results add a twist. Once we know a function is increasing, we can feed it two inputs whose order we already know (such as $x \leq \arcsin x$) and read off an inequality between the outputs. Spotting *which* two inputs to compare is the whole trick.

Solution

Part (i)(a)

Differentiating, using the product rule on $x\cos x$,

$$f'(x) = \cos x - (\cos x - x\sin x) = x\sin x.$$

For $0 \leq x \leq \frac{1}{2}\pi$ both $x \geq 0$ and $\sin x \geq 0$, so $f'(x) \geq 0$ and f is increasing on this interval.

Hence for $0 \leq x \leq \frac{1}{2}\pi$ we have $x \geq 0$, so $f(x) \geq f(0) = \sin 0 - 0 = 0$, as required.

Part (i)(b)

Let $h(x) = \arcsin x - x$ for $0 \leq x < 1$. Then

$$h'(x) = \frac{\mathrm{d}}{\mathrm{d}x}(\arcsin x) - 1 \geq 1 - 1 = 0,$$

using the given fact. So h is increasing, and $h(x) \geq h(0) = 0$. That is, $\arcsin x \geq x$ for $0 \leq x < 1$.

(The given fact is just $\dfrac{1}{\sqrt{1-x^2}} \geq 1$, true because $0 < \sqrt{1-x^2} \leq 1$.)

Part (i)(c)

Write $g(x) = \dfrac{x}{\sin x}$. For $0 < x < \frac{1}{2}\pi$, the quotient rule gives

$$g'(x) = \frac{\sin x - x\cos x}{\sin^2 x} = \frac{f(x)}{\sin^2 x}.$$

By part (a), $f(x) \geq 0$ here, and $\sin^2 x > 0$, so $g'(x) \geq 0$ and g is increasing on $(0, \frac{1}{2}\pi)$.

> **Ask yourself** How does knowing that g is increasing help with an inequality involving $\arcsin x$?

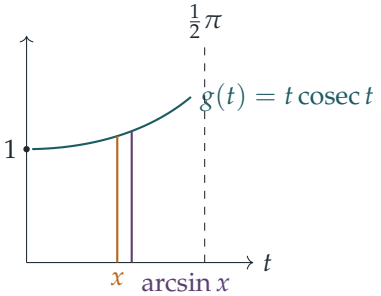
Look at the target: $\dfrac{\arcsin x}{x} \geq \dfrac{x}{\sin x}$. The right-hand side is $g(x)$. The left-hand side is also a value of g. If $y = \arcsin x$ then $\sin y = x$, so

$$g(y) = \frac{y}{\sin y} = \frac{\arcsin x}{x}.$$

So we need $g(\arcsin x) \geq g(x)$, and that follows from $\arcsin x \geq x$, which is part (b), because g is increasing.

To finish, take $0 < x < 1$ and let $y = \arcsin x$. Then $0 < y < \frac{1}{2}\pi$. Also $0 < x < 1 < \frac{1}{2}\pi$, so both x and y lie in the interval where g is increasing. By part (b), $y \geq x$, so $g(y) \geq g(x)$:

$$(\arcsin x)\, x^{-1} = \frac{y}{\sin y} \geq \frac{x}{\sin x} = x \operatorname{cosec} x \qquad (0 < x < 1).$$



The picture shows the argument for $x = 0.8$: the input $\arcsin x$ lies to the right of x, so the increasing curve is higher there.

Part (ii)

We copy the pattern of (i). First, the analogue of (b). Let $m(x) = x - \arctan x$ for $x \geq 0$. By the given fact, $m'(x) = 1 - \dfrac{\mathrm{d}}{\mathrm{d}x}(\arctan x) \geq 0$, so $m(x) \geq m(0) = 0$. Hence

$$\arctan x \leq x \qquad (x \geq 0).$$

Next, the analogue of (c). Let $k(x) = x^{-1}\tan x$ for $0 < x < \frac{1}{2}\pi$. Then

$$k'(x) = \frac{x\sec^2 x - \tan x}{x^2} = \frac{x - \sin x\cos x}{x^2\cos^2 x} = \frac{2x - \sin 2x}{2x^2\cos^2 x},$$

multiplying top and bottom by $\cos^2 x$ (which is positive) and then by 2.

? **Ask yourself** Why is $2x - \sin 2x \geq 0$?

This is the familiar $\sin t \leq t$ for $t \geq 0$, with $t = 2x$. It can be proved in the same style as everything else: $p(t) = t - \sin t$ has $p'(t) = 1 - \cos t \geq 0$ and $p(0) = 0$, so $p(t) \geq 0$ for $t \geq 0$.

So the numerator of $k'(x)$ is non-negative and the denominator is positive, giving $k'(x) \geq 0$. Thus k is increasing on $(0, \frac{1}{2}\pi)$.

Now take $0 < x < \frac{1}{2}\pi$ and let $y = \arctan x$. Then $0 < y < \frac{1}{2}\pi$, so both x and y lie in the interval where k is increasing, and $\tan y = x$. Since $y \leq x$, we get $k(y) \leq k(x)$:

$$\frac{x}{\arctan x} = \frac{\tan y}{y} \leq \frac{\tan x}{x}.$$

Both x and $\arctan x$ are positive, so multiplying through by $x\arctan x$ keeps the direction of the inequality:

$$(\tan x)(\arctan x) \geq x^2 \qquad (0 < x < \tfrac{1}{2}\pi),$$

as required.

Checks

- At $x = \frac{1}{2}$ in (c): $\arcsin(\frac{1}{2})/\frac{1}{2} = \pi/3 \approx 1.0472$, while $\frac{1}{2}\operatorname{cosec}\frac{1}{2} \approx 1.0429$. The inequality holds, and only just, which is what we expect: both sides tend to 1 as $x \to 0$.
- At $x = 1$ in (ii): $\tan 1 \cdot \arctan 1 = \frac{1}{4}\pi\tan 1 \approx 1.2232 \geq 1$.
- For small x, $\tan x \approx x + \frac{1}{3}x^3$ and $\arctan x \approx x - \frac{1}{3}x^3$, so the product is $x^2 + O(x^6)$: the inequality in (ii) is extremely tight near 0. A numerical check at 20,000 random points confirmed every inequality and every derivative formula above.

Common slips

- In (c), comparing g at the wrong pair of points, or forgetting to check that $\arcsin x$ is inside $(0, \frac{1}{2}\pi)$, where g is known to be increasing.
- In (ii), the inequality between the inputs points the other way ($\arctan x \leq x$), so the inequality between the outputs does too. Getting this backwards gives the wrong inequality.
- Multiplying an inequality through without saying that the multiplier ($x\arctan x$, or $\cos^2 x$) is positive.

Beyond STEP

The principle used throughout, that a non-negative derivative forces a function to increase, is a consequence of the mean value theorem, which is proved in any first course in analysis. The trick of comparing $g(x)$ with $g(\arcsin x)$, where sin and arcsin are inverses, is a cousin of ideas you'll meet with inverse functions and convexity. Abbott's *Understanding Analysis* gives a careful, readable proof of the mean value theorem and its consequences.

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